

Exercise 8 Backward in Space.

Given Equation (20):

$$U(x - \Delta x) = U(x) - \frac{\partial U}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 U}{\partial x^2} \Delta x^2 - \frac{1}{3!} \frac{\partial^3 U}{\partial x^3} \Delta x^3 + O(\Delta x^4)$$

First-Order approximation

$$U(x - \Delta x) \approx U(x) - \frac{\partial U}{\partial x} \Delta x$$

Re-arranging for velocity Gradient

$$U(x - \Delta x) \approx U(x) - \frac{\partial U}{\partial x} \Delta x$$

~~$$\frac{\partial U}{\partial x} \Delta x = U(x) - \frac{\partial U}{\partial x} \Delta x$$~~

$$\frac{\partial U}{\partial x} \Delta x \approx U(x) - U(x - \Delta x)$$

$$\frac{\partial U}{\partial x} \approx \frac{U(x) - U(x - \Delta x)}{\Delta x}$$

$$\frac{\partial U}{\partial x} = \frac{U(x) - U(x - \Delta x)}{\Delta x} + O(\Delta x)$$

Discrete approximation: $U_i = U(x_i)$

$$U_{i-1} = U(x_{i-1})$$

∴ The back up in space: $\left(\frac{\partial U}{\partial x} \right)_i \approx \frac{U_i - U_{i-1}}{\Delta x}$